

Detection of the Time-dependent Electromagnetic Potential at 1.3 GHz

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CEM-R-46

November 2007

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(Dated: November 2, 2007)

Abstract

We investigate the measurability of the 4-vector potential at radio frequencies (RF). We construct a wireless link where, ideally, the field vectors are zero but the electromagnetic 4-vector potential is not. The magnetic vector potential component of the 4-vector potential is longitudinally polarized with respect to the direction of propagation. Our experiments lead us to conclude that: (i) the RF 4-vector potential is measurable and can be detected with biased low-temperature plasma devices; and (ii) the RF power at the receiver is a result of conversion of the local bias power and is not due to RF power radiated by the transmitter.

PACS numbers: 03.50.De, 41.20.Jb, 52.35.Hr

Keywords: Electromagnetic potentials, Aharonov-Bohm effect, Low-temperature plasma, Classical electrodynamics

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I. INTRODUCTION

The physical significance and the measurability of the electromagnetic potential is a matter of long standing controversy. Experiments based on the Aharonov-Bohm effect [1–6] rely on electron wave interference to detect static potentials through the phase shifts they cause. A possible method to detect time-varying vector potential at optical frequencies is proposed in [7]. In most of these experiments, the interaction with the electron wave occurs in regions of space where the field vectors are practically zero but the potential is not. These experiments together with the intrinsic ability of the 4-vector potential to describe the momentum–energy state of an electromagnetic system is the reason why, in quantum electrodynamics, there is little if any doubt about the primary role of the potential as compared to that of the field vectors $\mathbf{E}$ and $\mathbf{H}$.

In contrast, in classical electrodynamics, the predominant opinion is that the 4-vector potential may be a convenient computational tool but it is not measurable and, therefore, carries no physical meaning. This opinion goes back to Heaviside [8] who discarded the potentials—introduced by Maxwell as the primary quantities—as nonphysical. Although Heaviside held Maxwell in high respect, he felt “obliged to dispense” with the “two potentials” for they rendered the equations of propagation “unmanageable and also not sufficiently comprehensive.” Heaviside (and Hertz independently) stated the “duplex” field equations (now known as Maxwell’s equations) and the associated four field vectors as the basis of electromagnetism. This view has come to disagree with quantum electrodynamics, making the modern science of electromagnetism unnaturally split into two branches with contradicting views on the basics.

There have been many reasons to regard the potentials as nonphysical quantities. First, there was no experimental evidence—until the confirmation of the Aharonov-Bohm effect by Chambers [2]—that there were any observable effects related to the electromagnetic potentials except indirectly through the field vectors.

Second, the nonuniqueness of the field representation in terms of potentials has been commonly regarded as nonphysical. This particular opinion deserves a closer look as it does not seem to have solid foundations. There exists an infinite set of field-invariant transformations of the electromagnetic potentials. This set includes the gauge transformations as a subset. Such transformations are well known and have been used to represent the field in terms of

two or even one scalar function of space and time (field scalarization, field decomposition), see, e.g. [9–15]. In [16], however, it is shown that this nonuniqueness is due to the field invariance to specific transformations of the electromagnetic sources. In other words, a given $(\mathbf{E}, \mathbf{H})$ field may result from any current-charge density 4-vector, which belongs to an infinite set of sources obtained from the original source through specific transformations. The field-invariant source transformations are analogous to the field-invariant potential transformations. For example, adding a gradient term to a current-density source does not change the resulting field outside of the source volume; however, it does change the vector and scalar components of the 4-potential in accordance with the gauge-invariant transformation. The invariance of the electromagnetic field to source transformations is well recognized in the solutions of inverse problems—they are intrinsically nonunique. And yet, this does not lead us to question the physical significance of the concepts of current and charge densities.

Third, in antenna theory, where the magnetic and electric vector potentials are common analytical tools, see, e.g., [10, 17], a number of paradoxes occur, in which the mathematical models predict nonzero propagating potentials with zero field vectors. We briefly note that the Lorenz gauge is common in these electrodynamic analyses. The most striking example is the expansion of the free-space field in spherical harmonics [10, 18] where the vector potential is radially polarized. The 0th-order solution, which features a spherically symmetric potential, appears to be nonphysical because the field vectors $\mathbf{E}$ and $\mathbf{H}$ are zero everywhere. At the same time, the wave impedance remains finite and exactly equal to that of free space. No energy can be coupled to this impedance, however, because the model implies a radially propagating “potential wave” with no power transport (the Poynting vector is zero). In general, a 4-vector potential $(\mathbf{A}, \Phi)$ results in zero field vectors in the far zone when $\mathbf{A}$ is purely longitudinal (polarized in the direction of propagation). Simple examples include the 0th-order spherical wave described above or the field along the axis of a dipole or a monopole.

Our work is an attempt to bring light to the discussion on the measurability of the 4-vector potential as it relates to classical electrodynamics and, more specifically, to radiation at RF/microwave frequencies. In the next section, we develop a theoretical model of the interaction of $(\mathbf{A}, \Phi)$ with moving charges in the nonrelativistic case. The model is based on the Lagrange equation of motion applied to low-temperature plasma and the solution of a boundary-value problem in a cylindrical region. It is tailored to our experimental setup

where fluorescent tubes and neon bulbs are used as detectors. In section III, we briefly outline the plasma model used to obtain needed plasma parameters such as electron number density and electron mobility. In section IV, we present some results based on our analytical model. Section V describes the detection scheme and the experimental set-up. The experimental results are discussed in section VI. We conclude with a summary and a discussion of the implications of these results.

II. THEORETICAL MODEL

In the following, we describe the assumptions made in modeling the generation of the 4-potential wave, its propagation, and its interaction with the low-temperature plasma detector.

A. The 4-potential of a Dipole

We assume the existence of a longitudinal vector potential due to external sources, which is polarized and propagating along the z -axis. For a monochromatic traveling wave, the vector potential in phasor notation is

$$\tilde{\mathbf{A}}^{ext} = \tilde{A}_z^{ext} \hat{\mathbf{z}}, \quad \tilde{A}_z^{ext} = A_m^{ext} e^{-jk_0 z}. \quad (1)$$

Here, $k_0 = \omega \sqrt{\epsilon_0 \mu_0}$ is the free-space wavenumber and ω is the radian frequency. The free-space permittivity and permeability are ϵ_0 and μ_0 , respectively. The factor $e^{j\omega t}$ is omitted throughout. The 4-vector potential $(\tilde{\mathbf{A}}^{ext}, \tilde{\Phi}^{ext})$ is considered in the far zone of its sources and, therefore, the potential wave is assumed locally-plane, i.e., $\partial \tilde{\mathbf{A}}^{ext} / \partial x = \partial \tilde{\mathbf{A}}^{ext} / \partial y = 0$. The magnetic flux density vector $\tilde{\mathbf{B}}^{ext} = \nabla \times \tilde{\mathbf{A}}^{ext}$ of such a wave is zero since $\tilde{\mathbf{A}}^{ext}$ is locally curl-free. The electric field $\tilde{\mathbf{E}}^{ext}$,

$$\tilde{\mathbf{E}}^{ext} = -j\omega \tilde{\mathbf{A}}^{ext} - \nabla \tilde{\Phi}^{ext}, \quad (2)$$

is zero as well provided $\tilde{\Phi}^{ext}$ satisfies the Lorenz gauge:

$$j\omega\epsilon_0\mu_0\tilde{\Phi}^{ext} = -\frac{\partial \tilde{A}_z^{ext}}{\partial z}. \quad (3)$$

The above implies that the density source of $(\tilde{\mathbf{A}}^{ext}, \tilde{\Phi}^{ext})$ obeys the continuity relation and is z -polarized. Since $\partial \tilde{A}_z^{ext} / \partial z = -jk_0 \tilde{A}_z^{ext}$,

$$\tilde{\Phi}^{ext} = c \tilde{A}_z^{ext}, \quad (4)$$

where $c = (\mu_0 \epsilon_0)^{-\frac{1}{2}}$ is the speed of light in vacuum. This scenario describes closely the far-zone field along the axis of a dipole or a monopole.

The far-zone $\tilde{\mathbf{A}}^{ext}$ due to a dipole of length $L^{(d)}$ oriented along the z -axis can be obtained from [17]

$$\tilde{\mathbf{A}}_z^{ext}(P) = \hat{\mathbf{z}} \frac{\mu_0}{4\pi} \cdot \frac{e^{-jk_0 R_0}}{R_0} \cdot \int_{-L^{(d)}/2}^{L^{(d)}/2} I(z') e^{jk_0 z' \cos \theta} dz', \quad (5)$$

where R_0 is the distance from the antenna center to the point of observation P and θ is the observation angle in the elevation plane. The potential is independent of the azimuthal angle φ due to the cylindrical symmetry of the source. The current distribution along the wire $I(z')$ is approximated well by the sinusoidal distribution

$$I_m \sin [k_0(L^{(m)} - |z'|)] , -L^{(m)} \leq z' \leq L^{(m)}. \quad (6)$$

Here, $L^{(m)}$ is half the length of the dipole, which is also the length of the equivalent monopole, $L^{(m)} = 0.5L^{(d)}$. The current's magnitude is denoted as I_m .

With this current distribution, the far-zone potential integral Eq. (5) has analytical solution whose normalized pattern is given by

$$\bar{A}_z^{ext}(\theta) = \frac{\cos(L_e^{(m)} \cos \theta) - \cos L_e^{(m)}}{1 - \cos^2 \theta}, \quad (7)$$

where $L_e^{(m)} = k_0 L^{(m)}$ is the electrical half-length of the dipole. At $\theta = 0$, the pattern value is $0.5L_e^{(m)} \sin(L_e^{(m)})$. To obtain the actual magnitude of the far-zone A_z^{ext} , it suffices to multiply the result in Eq. (7) by

$$A_m = \frac{\mu_0}{2\pi} \cdot \frac{I_m}{k_0 R_0}. \quad (8)$$

The 3-D normalized pattern $\bar{A}_z^{ext}$ for a half-wavelength dipole is plotted in Fig. 1 together with its 2-D elevation cut. The pattern is fairly uniform with the potential magnitude varying between $\frac{\pi}{4}$ at $\theta = 0$ and 1.0 at $\theta = 90^\circ$. The shorter the dipole is the more uniform its potential pattern becomes. There is significant difference between the potential pattern and the ‘donut’ shaped field-magnitude pattern, which has a null along the dipole’s axis and a maximum in the azimuthal plane.

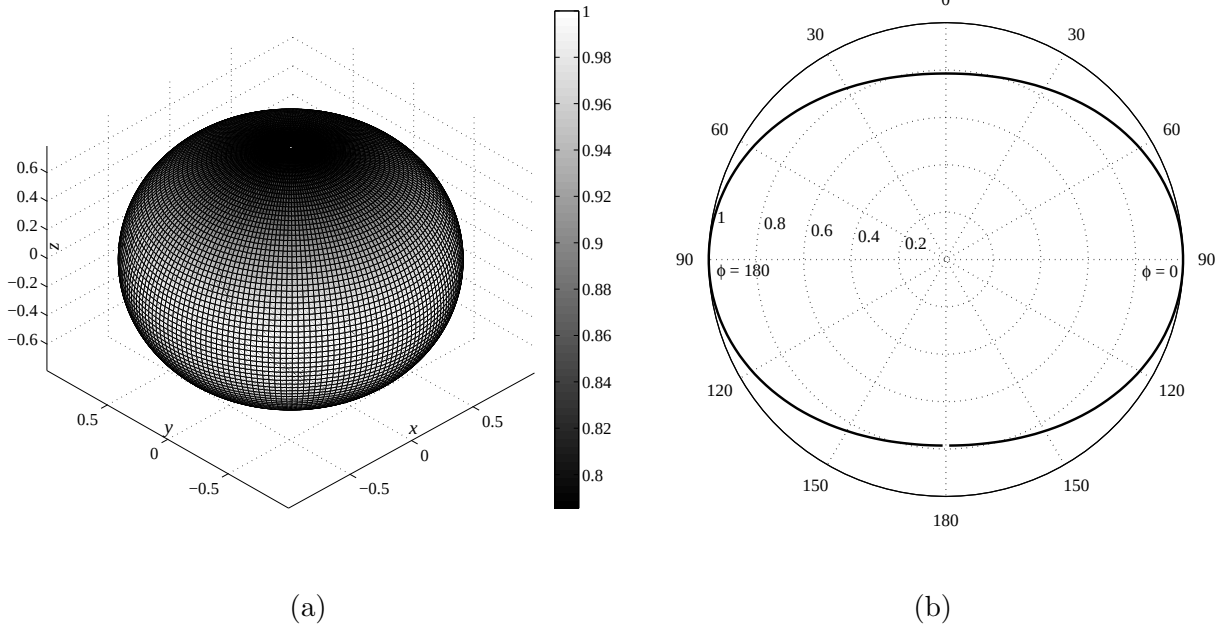


FIG. 1: Patterns of the far-zone magnetic vector potential A_z of a half-wavelength dipole: (a) 3-D pattern; (b) 2-D pattern in the elevation plane.

In summary, a longitudinal (A_z, Φ) potential wave is generated along the axis of a dipole or a monopole within a narrow conical region where the associated $(\mathbf{E}, \mathbf{H})$ field is much suppressed.

B. The Equation of Motion

We consider low-temperature (cold) plasma as the medium interacting with the 4-potential. The high-frequency electromagnetic response of the plasma is mostly due to the electrons whose mass is much less than that of the ions. Neglecting the pressure term, the motion of the electrons is described by the nonrelativistic equation [19]

$$\frac{d\mathbf{v}}{dt} = R_e (\mathbf{E} + \mathbf{v} \times \mathbf{B}) , \quad (9)$$

where $R_e = \frac{e}{m_e}$ is the charge-to-mass ratio of the electron, $\mathbf{v}$ is its velocity, and $\frac{d\mathbf{v}}{dt}$ is the total acceleration. Taking loss into account [20], the total acceleration is decomposed as

$$\frac{d\mathbf{v}}{dt} = \frac{\partial^e \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} , \quad (10)$$

where, the term $\frac{\partial^e \mathbf{v}}{\partial t} = \frac{\partial \mathbf{v}}{\partial t} + \frac{\mathbf{v}}{\tau}$ gives the explicit dependence of the acceleration on time. Its second component represents loss. The parameter τ (average momentum relaxation time) relates to the effective plasma DC conductivity σ_{DC} as

$$\tau = \frac{\sigma_{DC}}{R_e \rho_0}, \quad (11)$$

where $\rho_0 = en_0$ is the charge volume density and n_0 is the electron number density. These densities are assumed time and space invariant on average. In plasma description, the parameter collision frequency $\nu_{en} = \tau^{-1}$ is sometimes preferred. In cold plasma, the electron collisions with neutral particles (atoms) largely determine the DC conductivity. We briefly note that the loss mechanism in discharge Ne or Ar-Hg plasma is generally very complicated—it includes different types of collisions, wall diffusion, recombination and radiation in the visible and the UV spectrum. Later, a simplified model will be presented, which is based on empirical data from the literature and DC measurements of fluorescent tubes used in our experiments. It allows for the approximate calculation of n_0 and ν_{en} .

Substituting Eq. (10) in Eq. (9) and applying standard vector identities [21], we write Eq. (9) as

$$\frac{\partial \mathbf{v}}{\partial t} + \frac{\mathbf{v}}{\tau} + \nabla \left(\frac{\mathbf{v} \cdot \mathbf{v}}{2} \right) - R_e \mathbf{E} = \mathbf{v} \times (\nabla \times \mathbf{v} + R_e \mathbf{B}), \quad (12)$$

Eq. (12) is the equation of viscous electronic fluid in an external electromagnetic field. In some plasma models, the acceleration term $\frac{\partial \mathbf{v}}{\partial t}$ in Eq. (9) is neglected. In our case, however, the electron-neutral collision frequency is of the same order as the frequency of the electromagnetic wave ($\nu_{en} \approx 1.4 \times 10^9 \text{ s}^{-1}$) and, therefore, the acceleration term is significant—it is similar in magnitude to the loss term.

After the field vectors are expressed in terms of the vector and scalar potentials, Eq. (12) becomes

$$\frac{\partial}{\partial t}(\mathbf{v} + R_e \mathbf{A}) + \frac{\mathbf{v}}{\tau} + \nabla \left(\frac{\mathbf{v} \cdot \mathbf{v}}{2} + R_e \Phi \right) = \mathbf{v} \times \nabla \times (\mathbf{v} + R_e \mathbf{A}). \quad (13)$$

An equation relating only $\mathbf{v}$ and $\mathbf{A}$ is obtained by taking the curl of both sides of Eq (13):

$$\frac{\partial \mathbf{w}}{\partial t} + \nabla \times \left(\frac{\mathbf{v}}{\tau} - \mathbf{v} \times \mathbf{w} \right) = 0, \quad (14)$$

where $\mathbf{w} = \nabla \times (\mathbf{v} + R_e \mathbf{A})$. Following the reasoning in [21], if at some initial state of the system

$$\mathbf{w} = 0 \text{ and } \nabla \times \mathbf{v} = 0 \text{ (at } t = 0), \quad (15)$$

then from Eq. (14) it follows that

$$\frac{\partial \mathbf{w}}{\partial t} = 0 \text{ (at } t = 0 \text{)}. \quad (16)$$

The initial conditions in Eq. (15) could be realized, for example, by $\mathbf{v} = 0$ and $\mathbf{A} = 0$ (no external field) at $t = 0$, or by $\nabla \times \mathbf{v} = 0$ (direct current due to static voltage or current source) and $\mathbf{A} = 0$ at $t = 0$, etc. The last result in Eq. (16) means that $\mathbf{w}$ is zero at any time in a system where it was initially zero. This is expected in a homogeneous, i.e., source-free, equation such as Eq. (14).

In a system where $\mathbf{w} = 0$, Eq. (13) simplifies to

$$\frac{\partial}{\partial t}(\mathbf{v} + R_e \mathbf{A}) + \frac{\mathbf{v}}{\tau} + \nabla \left(\frac{\mathbf{v} \cdot \mathbf{v}}{2} + R_e \Phi \right) = 0, \quad (17)$$

which is equivalent to

$$R_e \mathbf{E} = \frac{\partial \mathbf{v}}{\partial t} + \frac{\mathbf{v}}{\tau} + \nabla \left(\frac{\mathbf{v} \cdot \mathbf{v}}{2} \right). \quad (18)$$

Eq. (17) states the relation between the canonical momentum and the energy of an electron when $\mathbf{w} = 0$.

C. Time-harmonic Solution in a Cylindrical Infinitely Long Region

We next consider plasma confined within a cylindrical region of radius a and oriented along z . The plasma is DC biased and the electrons are initially set into motion with a constant velocity $\mathbf{v}_0$ averaged over the plasma volume. This velocity is assumed uniform throughout the plasma region. The incident field is a longitudinal potential wave where $\tilde{\mathbf{A}}^{ext} = \tilde{A}_z^{ext} \hat{\mathbf{z}}$ and $\tilde{\Phi}^{ext} = c \tilde{A}_z^{ext}$, as discussed in section II A. This wave propagates along z and is locally plane. It has zero $\mathbf{E}^{ext}$ and $\mathbf{H}^{ext}$ field vectors. The solution for all velocity and potential variables is sought in the form $\sim \exp(-jk_0 z)$ so that the propagation mode in the cylinder can couple to the incident wave.

1. Time-harmonic Formalism

The interaction of the electrons with the incident wave is represented by Eq. (17) written for the first-order harmonic terms:

$$j\omega \tilde{\mathbf{v}} + \frac{\tilde{\mathbf{v}}}{\tau} + \nabla(\boldsymbol{\nu} \cdot \tilde{\mathbf{v}}) = -R_e(j\omega \tilde{\mathbf{A}} + \nabla \tilde{\Phi}), \quad (19)$$

where $\tilde{\mathbf{v}}$ is the first-order harmonic variation of the electron velocity. Due to the first-order harmonic nature of the term $(\boldsymbol{\nu} \cdot \tilde{\mathbf{v}})$, the velocity vector $\boldsymbol{\nu}$ is not a wave phasor and does not depend on position and time. It is not the averaged particle velocity $\mathbf{v}_0$ —the velocity disturbance $\tilde{\mathbf{v}}$ is a wave moving at the speed of light and its interaction with a slowly moving particle is negligible. In the case of a particle continuum, however, it can interact with the charge disturbance $\tilde{\rho}$ which results from $\tilde{\mathbf{v}}$ as per the continuity law and which propagates at the speed of light itself. The vector $\boldsymbol{\nu}$ is determined later.

Eq. (19) alone does not provide sufficient information to find the explicit relations between the components of the 4-vector current density, the 4-vector potential, $\tilde{\mathbf{v}}$ and $(\boldsymbol{\nu} \cdot \tilde{\mathbf{v}})$. We will additionally require that the components of the potential satisfy the Lorenz gauge,

$$\nabla \cdot \tilde{\mathbf{A}} = -\frac{j\omega}{c^2} \tilde{\Phi}. \quad (20)$$

Here, it is appropriate to emphasize that arbitrary choices of the gauge condition, e.g., $\nabla \cdot \tilde{\mathbf{A}} = 0$, are contrary to any attempt to search for the physical significance of the electromagnetic potentials. They may preserve the field components intact [22], however, they imply changes in the source 4-vector as discussed in the Introduction. As Lorenz showed [23–25], the gauge he proposed is a mathematical consequence of the definition of the magnetic vector potential and the electric scalar potential as the retardation integrals over current and charge sources, respectively, as well as the continuity relation between these sources. The Lorenz gauge is a choice with solid foundation.

Making use of Eq. (19) and Eq. (20), we arrive at the relation between the velocity and potential components:

$$-R_e \tilde{\mathbf{A}} = \eta \tilde{\mathbf{v}}, \quad (21)$$

$$-R_e \tilde{\Phi} = \boldsymbol{\nu} \cdot \tilde{\mathbf{v}}, \quad (22)$$

as well as the relation between the induced sources and the 4-vector potential:

$$\tilde{\mathbf{J}} = -\frac{\rho_0 R_e}{\eta} \cdot \tilde{\mathbf{A}}, \quad (23)$$

$$\tilde{\rho} = -\frac{\rho_0 R_e}{c^2 \eta} \cdot \tilde{\Phi}, \quad (24)$$

where $\eta = 1 + \frac{1}{j\omega\tau}$. The derivation is detailed in the Appendix.

2. Solution in Cylindrical Coordinates

From Maxwell's equations in vacuum and making use of the Lorenz gauge, the Helmholtz equations for the potentials in the presence of sources are obtained:

$$\nabla^2 \tilde{\mathbf{A}} + k_0^2 \tilde{\mathbf{A}} = -\mu_0 \tilde{\mathbf{J}}, \quad (25)$$

$$\nabla^2 \tilde{\Phi} + k_0^2 \tilde{\Phi} = -\frac{\tilde{\rho}}{\epsilon_0}. \quad (26)$$

Substituting Eq. (23) and Eq. (24) in the above equations, we obtain the governing equations in the plasma medium:

$$\nabla^2 \tilde{\mathbf{A}} + (k_0^2 - p^2) \tilde{\mathbf{A}} = 0, \quad (27)$$

$$\nabla^2 \tilde{\Phi} + (k_0^2 - p^2) \tilde{\Phi} = 0, \quad (28)$$

where

$$p^2 = \frac{R_e \mu_0 \rho_0}{\eta}. \quad (29)$$

Further, it follows from Eq. (21) and Eq. (22) that $\tilde{\mathbf{v}}$ and $\tilde{\mathbf{J}}$ satisfy the same equation as Eq. (27), as well as $(\boldsymbol{\nu} \cdot \tilde{\mathbf{v}})$ and $\tilde{\rho}$ satisfy the same equation as Eq. (28).

In solving Eq. (27) and Eq. (28), we assume that the cylindrical volume is symmetric with respect to the angular coordinate φ and so is the external wave. Making use of $\frac{\partial^2}{\partial z^2} = -k_0^2$ and $\frac{\partial}{\partial \varphi} = 0$, the Helmholtz equations in cylindrical coordinates become

$$\frac{\partial^2 \tilde{A}_r^{(1)}}{\partial r^2} + \frac{1}{r} \frac{\partial \tilde{A}_r^{(1)}}{\partial r} - \frac{\tilde{A}_r^{(1)}}{r^2} - p^2 \tilde{A}_r^{(1)} = 0, \quad (30)$$

$$\frac{\partial^2 \tilde{A}_z^{(1)}}{\partial r^2} + \frac{1}{r} \frac{\partial \tilde{A}_z^{(1)}}{\partial r} - p^2 \tilde{A}_z^{(1)} = 0, \quad (31)$$

$$\frac{\partial^2 \tilde{\Phi}^{(1)}}{\partial r^2} + \frac{1}{r} \frac{\partial \tilde{\Phi}^{(1)}}{\partial r} - p^2 \tilde{\Phi}^{(1)} = 0. \quad (32)$$

Here, we have introduced the superscript (1) in order to indicate the plasma volume as region 1 in distinction to the infinite air region surrounding it, which will be indicated as region 2.

In air, for the lack of induced sources, the potential equations are

$$\frac{\partial^2 \tilde{A}_r^{(2)}}{\partial r^2} + \frac{1}{r} \frac{\partial \tilde{A}_r^{(2)}}{\partial r} - \frac{\tilde{A}_r^{(2)}}{r^2} = 0, \quad (33)$$

$$\frac{\partial^2 \tilde{A}_z^{(2)}}{\partial r^2} + \frac{1}{r} \frac{\partial \tilde{A}_z^{(2)}}{\partial r} = 0, \quad (34)$$

$$\frac{\partial^2 \tilde{\Phi}^{(2)}}{\partial r^2} + \frac{1}{r} \frac{\partial \tilde{\Phi}^{(2)}}{\partial r} = 0. \quad (35)$$

Note that $\tilde{A}_z$ and $\tilde{\Phi}$ satisfy the same equation in each region.

The general solutions in regions 1 and 2 are

$$\tilde{A}_z^{(1)}(r) = C_1 I_0(pr), \quad (36)$$

$$\tilde{A}_r^{(1)}(r) = C_2 I_1(pr), \quad (37)$$

$$\tilde{\Phi}^{(1)}(r) = \zeta_1 I_0(pr), \quad (38)$$

and

$$\tilde{A}_z^{(2)}(r) = C_3 \ln\left(\frac{\kappa}{r}\right) + C_4, \quad (39)$$

$$\tilde{A}_r^{(2)}(r) = \frac{C_5}{r}, \quad (40)$$

$$\tilde{\Phi}^{(2)}(r) = \zeta_2 \ln\left(\frac{\kappa}{r}\right) + \zeta_3. \quad (41)$$

Here, C_i ($i = 1, \dots, 5$), ζ_i ($i = 1, 2, 3$) and κ are integration constants. I_0 and I_1 are the modified Bessel function of order zero and one, respectively. The other modified Bessel functions K_0 and K_1 , which solve the equations in region 1, are not suitable because they are infinite at $r = 0$.

The general solution for the field components is obtained from the potential solution as

$$\tilde{H}_\varphi^{(1)} = -\mu_0^{-1}(pC_1 + jk_0C_2)I_1(pr), \quad (42)$$

$$\tilde{E}_z^{(1)} = -cpC_2I_0(pr), \quad (43)$$

$$\tilde{E}_r^{(1)} = -[j\omega(1 + c^2p^2\omega^{-2})C_2 + cpC_1]I_1(pr), \quad (44)$$

and

$$\tilde{H}_\varphi^{(2)} = -\mu_0^{-1}(jk_0C_5 - C_3)\frac{1}{r}, \quad (45)$$

$$\tilde{E}_z^{(2)} = 0, \quad (46)$$

$$\tilde{E}_r^{(2)} = (cC_3 - j\omega C_5)\frac{1}{r}. \quad (47)$$

The integration constants are determined as follows. The continuity of E_z requires that $C_2 = 0$, which sets $\tilde{A}_r^{(1)} = 0$. By the continuity of the potential r -component, $C_5 = 0$ and $\tilde{A}_r^{(2)} = 0$. With a vanishing r -component of the vector potential, the Lorenz gauge requires that in both regions

$$\tilde{\Phi}^{(i)} = c\tilde{A}_z^{(i)}, \quad i = 1, 2. \quad (48)$$

Thus, hereafter, we may consider only $\tilde{A}_z$ and the remaining constants C_1 , C_3 , C_4 and κ .

The constant term in $\tilde{A}_z^{(2)}$ must represent the incident potential $\tilde{A}_z^{ext}$ which is assumed to be a uniform plane wave locally; thus, for $z = 0$,

$$C_4 = A_m^{ext}. \quad (49)$$

Further, the continuity of $\tilde{A}_z$ and $\tilde{H}_\varphi$ requires that

$$C_1 I_0(pa) = C_3 \ln(\kappa/a) + A_m^{ext}, \quad (50)$$

$$C_1 p I_1(pa) = -C_3/a. \quad (51)$$

The logarithmic term in the expression for $\tilde{A}_z^{(2)}$ in Eq. (39) is due to the induced current in the cylindrical plasma region, which is assumed infinitely long. This term is equivalent to the magnetostatic solution for the vector potential in a 2-D problem in the plane transverse to z for the case of a long filament with current. In reality, the plasma column in a fluorescent tube is of finite length L and the logarithmic term is valid only for $r \ll L$. To obtain κ and C_3 , we use the magnetostatic solution for the vector potential at an observation point lying in a plane which intersect in the middle a filament of length L and current I :

$$A_z^{st}(r) = \frac{\mu_0 I}{2\pi} \ln \left[\frac{L}{2r} + \sqrt{1 + \left(\frac{L}{2r} \right)^2} \right]. \quad (52)$$

The details of the integral solution can be found in [26]. If $L \gg r$, the above solution tends to

$$A_z^{st}(r) = \frac{\mu_0 I}{2\pi} \ln \left(\frac{L}{r} \right). \quad (53)$$

Comparing Eq. (53) with Eq. (50), we find that $\kappa = L$ and C_3 must be proportional to the total induced RF current $\tilde{I}_t$

$$C_3 = \frac{\mu_0 \tilde{I}_t}{2\pi}. \quad (54)$$

At the same time, the current density $\tilde{J}_z$ in the plasma volume can be calculated as a function of r from $\tilde{A}_z^{(1)}$ using Eq. (23):

$$\tilde{J}_z = -\frac{p^2}{\mu_0} \tilde{A}_z^{(1)}. \quad (55)$$

From $\tilde{J}_z$, the total current $\tilde{I}_t$ is obtained by integrating over the cross-section of the cylinder, which results in

$$\tilde{I}_t = -\frac{2\pi}{\mu_0} C_1 p a I_1(pa). \quad (56)$$

Substituting Eq. (56) into Eq. (54) results in a relation identical to Eq. (51), which confirms the validity of the magnetostatic analysis for the logarithmic term in Eq. (50). We can now express C_1 as

$$C_1 = \frac{A_m^{ext}}{I_0(pa) + paI_1(pa) \ln(L/a)}. \quad (57)$$

C_3 can be calculated from C_1 using Eq. (51).

It is important to note that the current component, which is in phase quadrature with $\tilde{A}_z^{(1)}$, relates to loss in the plasma region as per Eq. (21). If the tube is connected to an external RF load, the current component in phase with $\tilde{A}_z^{(1)}$, is the one which can deliver power. This current component is

$$\tilde{I} = \frac{\text{Re}(p^2)}{p^2} \tilde{I}_t, \quad (58)$$

where Re denotes the real part of the complex argument.

3. Consistency of the Solution

The consistency of the solution described above is contingent upon the fulfilment of all underlying relations between the 4-vector potential and the velocity terms. In particular, we observe that the relations containing the $\boldsymbol{\nu}$ vector (see Eq. (77), Eq. (79), and Eq. (81) in the Appendix) are consistent with the obtained solution only for a particular value of the $\boldsymbol{\nu}$ vector:

$$\boldsymbol{\nu} = c\eta\hat{\mathbf{z}}, \quad \eta = \left(1 + \frac{1}{j\omega\tau}\right). \quad (59)$$

It is then that the Lorenz gauge, $\tilde{\Phi}^{(1)} = c\tilde{A}_z^{(1)}$, and the continuity relation of the wave-current, $\tilde{J}_z = c\tilde{\rho}$, are fulfilled.

The same result can be obtained directly from Eq. (19) for the case of a vanishing longitudinal electric field component, $\tilde{E}_z = 0$:

$$j\omega\eta\tilde{v}_z - jk_0\nu\tilde{v}_z = 0, \quad (60)$$

which is fulfilled for non-zero $\tilde{v}_z$ only if Eq. (59) holds. Note that the case of $\tilde{E}_z = 0$ corresponds to the solution described in the previous section where nonzero current-density proportional to $\tilde{A}_z^{(1)}$ is obtained. Incidentally, Eq. (59) and Eq. (60) correspond to zero total (or co-moving) acceleration of the charge distribution along z .

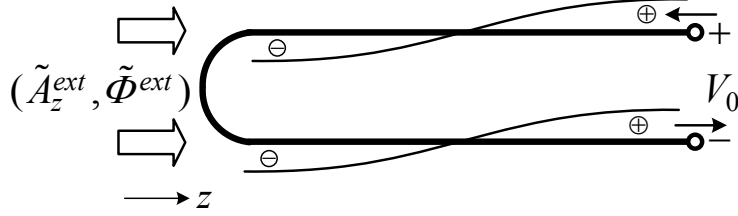


FIG. 2: Sketch of a U-shaped arrangement of two parallel plasma branches and the current due to the charge change as a result of the interaction with the incident wave at some initial moment of time.

We emphasize that an analysis based on the field vectors alone suggests only a trivial solution for $\tilde{v}_z$ and $\tilde{I}$. However, such analysis imposes more restrictions than implied by Eq. (60). The potential-based analysis outlined above leads to a possible nonzero solution contingent upon the fulfilment of Eq. (59).

It is clear that the incident wave—which carries no power and whose field vectors are zero—cannot provide the energy necessary to sustain the velocity fluctuations of the electrons, especially in a lossy medium. The current can be sustained only by the fluctuations of the term $\nabla(\boldsymbol{\nu} \cdot \tilde{\mathbf{v}})$ in Eq. (19) or $-jk_0\nu\tilde{v}_z$ in Eq. (60). This term, which is proportional to $\nabla\tilde{\rho}$, represents available energy in the system due to abundant supply of charge provided by the DC source.

It is also important to realize that the analysis in the previous section does not take into account the interaction of the DC source with the plasma, neither does it impose the condition of charge neutrality in the plasma according to which the total RF charge density must be zero, $\tilde{\rho} = 0$. Charge neutrality is maintained by the plasma medium if the operating frequency is well below the plasma frequency [27]. As we show in the next section, this is indeed the case in our experiments. In fact, a solution can be obtained for the case of zero charge density, which also means zero scalar potential, $\tilde{\Phi} = 0$, as per Eq. (24). This solution, however, implies a discontinuity of the scalar potential at the interaction boundary, which translates into a non-zero longitudinal $\tilde{E}_z^{ext}$ field component. With this field component imposed as a boundary condition, the final result is identical to the one obtained in the previous section. The field $\tilde{E}_z^{ext}$ can be viewed as due to the biased plasma, which acts to restore the uniform charge distribution.

In summary, the plasma medium remains charge-neutral, however, this is at the expense of charge injected by the DC source. The latter implies non-zero current. Thus, the plasma can be viewed as a time-varying load driven by the DC source, which results in respective time-varying current.

In Fig. 2, we sketch a possible arrangement which allows for an experimental verification. Two parallel branches (plasma columns) are biased in series. The interaction of the plasma with the incident wave in both tubes is identical and the phase of the induced current is in general arbitrary. It will entirely depend on the initial conditions of charge injection at the bias terminals. Since the two tubes are connected at terminals of opposite polarity, at a given initial moment in time, the identical charge change at both terminals will result in opposite currents at the two terminals as illustrated in Fig. 2. Thus, we can expect the induced wave-currents in both branches to be of opposite polarity and to propagate along z as required by Eq. (59). The propagation is identical to the familiar wave-current in a twin-lead transmission line.

It is our purpose to investigate experimentally the above theory.

III. NUMERICAL PLASMA MODEL

The theoretical model presented above depends on the eigenvalue p in Eq. (29), for which we need the following plasma parameters: (i) the electron number density n_0 , which yields the charge density $\rho_0 = en_0$, and (ii) the plasma electron mobility μ_e , which yields the momentum relaxation constant τ , or, the DC conductivity σ_{DC} , see Eq. (11). We briefly note that the charge density in a low-pressure discharge lamp, such as an Ar-Hg fluorescent lamp, is not constant with the radial distance from the tube axis as assumed in the theoretical model above. The simplest model assumes a density distribution which is a 0th-order Bessel function J_0 of the radial distance; see, e.g. [28, 29]. We use this dependence to determine an effective plasma column radius a_{eff} from the tube radius a as

$$a_{\text{eff}} = a \sqrt{\frac{2J_1(x_{01})}{x_{01}}}, \quad (61)$$

where $x_{01} = 2.4048$ is the 1st zero of $J_0(x)$ and $J_1(x)$ is the 1st-order Bessel function.

Plasma modeling is an extremely complex problem [30], which is currently under intense study. Here, we aim at approximate modeling of the plasma parameters based on some

measured input parameters and some assumed parameters as explained below. Our model yields the parameters n_0 and μ_e through the numerical solution of the nonlinear gas temperature equation under DC conditions. It is based on the balance of the heat generated by the electron-neutral particle elastic collisions and the heat removed by the gas thermal conduction [29]. The gas temperature equation is solved by the MATLAB function **fzero**.

The following input parameters must be supplied: 1) the measured tube wall temperature T_w after heat balance is achieved, 2) the measured DC voltage V_0 and current I_0 of the tube at the operating point, 3) the Ar gas pressure, and 4) the electron temperature. T_w depends on the DC bias, which translates into the dependence of n_0 and μ_e . We have measured the wall temperatures of a compact fluorescent U-shaped tube made by BAYCO (SL-213) at bias currents $I_0 = 100$ mA ($V_0 = 72.6$ V) and $I_0 = 200$ mA ($V_0 = 57.4$ V). At $I_0 = 100$ mA, $T_w = 40.8^\circ$ C. At $I_0 = 200$ mA, $T_w = 58.5^\circ$ C. The Ar gas pressure is set at $P_{\text{Ar}} = 3$ torr (≈ 400 Pa) as per available data in the literature, see, e.g., [31]. The typical electron temperature T_e in low-pressure DC Ar-Hg discharge, is about $11\,600^\circ$ K [31]. Thus, the electron temperature here is set equal to 1 eV, i.e., $kT_e = 1$ eV, where $k = 1.3806505 \times 10^{-23}$ J/K is Boltzmann's constant. We note that the data in the literature is for large fluorescent tubes only and may be somewhat different in compact tubes. Therefore, all modeling results presented hereafter are only approximate.

The iterative solution is summarized below.

1. Give initial value to the absolute gas temperature T_g in degrees Kelvin. A reasonable guess is a few degrees above the wall (cold-spot) temperature T_w , although the initial guess is not critical.
2. Determine the Hg vapor pressure from the empirical formula [29]

$$\ln(P_{\text{Hg}}(\text{mTorr})) = 30.804 - 0.8254 \ln(T_w) - \frac{7564}{T_w}. \quad (62)$$

T_w is in degrees Kelvin.

3. Determine the number density of Ar atoms:

$$n_{\text{Ar}} = \frac{P_{\text{Ar}}}{kT_g}. \quad (63)$$

4. Determine the number density of Hg atoms:

$$n_{\text{Hg}} = \frac{P_{\text{Hg}}}{kT_g}. \quad (64)$$

5. Compute numerically the electron-mobility integral [29]

$$\mu_e = -\frac{1}{3} \left(\frac{2e}{m_e} \right)^{1/2} \int_0^\infty \frac{u}{n_{\text{Ar}} Q_{\text{Ar}}^m + n_{\text{Hg}} Q_{\text{Hg}}^m} \cdot \frac{d}{du} \left(\frac{f(u)}{\sqrt{u}} \right) du, \quad (65)$$

where $e = 1.6021 \times 10^{-19}$ C is the electron charge, $m_e = 9.106 \times 10^{-31}$ kg is the electron mass, u is the electron energy, $f(u)$ is the electron energy distribution function (EEDF), Q_{Ar}^m and Q_{Hg}^m are the momentum transfer cross-sections of Ar and Hg, respectively. Plots of the dependence of Q_{Ar}^m and Q_{Hg}^m on the electron energy can be found in [27] and [29]. We have used the EEDF described by the Maxwell-Boltzmann equation

$$f(u) = 2 \left(\frac{u}{\pi} \right)^{1/2} (kT_e)^{-3/2} \exp \left(-\frac{u}{kT_e} \right), \quad (66)$$

where $kT_e = 1$ eV. We briefly note that more accurate results can be obtained by using the EEDF suggested by Maya and Lagushenko [32] specifically for the case of low-pressure Ar-Hg fluorescent lamps. However, this will complicate the numerical computation while our purpose at the present stage is to obtain predictions of accuracy within an order of magnitude.

6. Compute electron-neutral collision frequency [27]:

$$\nu_{en} = \frac{e}{m_e \mu_e}. \quad (67)$$

7. Compute axial electron number density n_0 using I_0 and V_0 :

$$n_0 = \frac{0.2}{e\mu_e} \cdot \frac{I_0}{\pi a_{\text{eff}}^2} \cdot \frac{L_t}{V_0}, \quad (68)$$

where a_{eff} is calculated using Eq. (61). The physical radius of the lamp's cross-section is $a = 5$ mm. The total folded-tube length is $L_t = 27.5$ cm along the geometrical mean path. The factor of 0.2 accounts for the portion of the loss power attributed to collisions. According to Lister [31], this portion is about 20% in most Ar-Hg fluorescent lamps, the rest being spent on electrode loss, wall diffusion, visible and UV radiation.

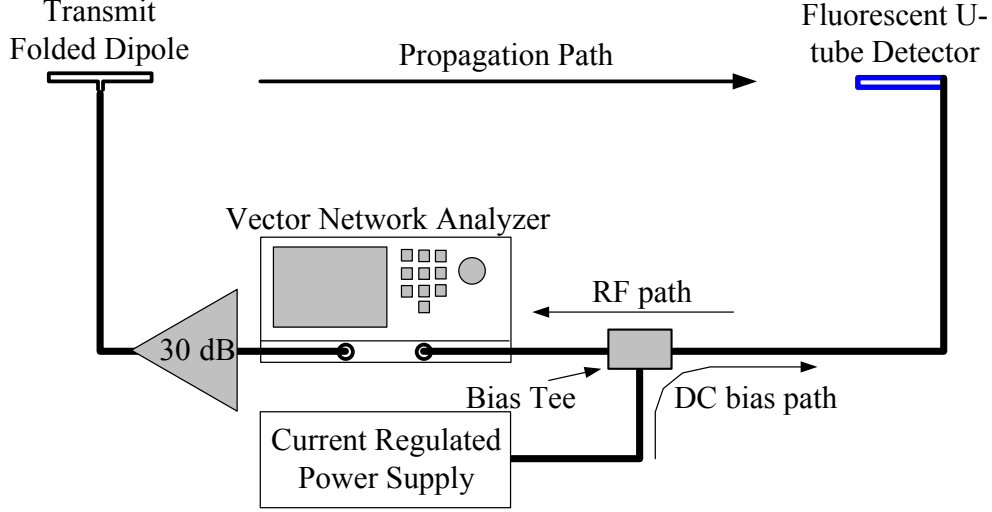


FIG. 3: Measurement setup for the detection of the potential.

8. Check gas-temperature balance [29]:

$$f(T_g) = \nu_{en} \frac{3m_e}{m_n} k(T_e - T_g) n_e - \frac{4}{a^2} \kappa(T_g - T_w) = 0, \quad (69)$$

where m_n is the mass of the Ar atom ($m_n = w_{\text{Ar}} \times 10^{-3}/A$ kg, $w_{\text{Ar}} = 39.948$ is the Ar atomic weight, $A = 6.0221415 \times 10^{23}$ is Avogadro's number). The thermal conductivity of the Ar gas is taken as $\kappa = 4.17 \times 10^{-4} \cdot T_g^{2/3}$ [29].

If $f(T_g) = 0$, stop; else, assign new value to T_g and proceed to step 3.

The above numerical algorithm yields the following plasma parameters at $I_0 = 200$ mA: gas temperature $T_g = 59.39^\circ$ C, electron number density $n_0 = 2.981 \times 10^{17} \text{ m}^{-3}$, electron-neutral collision frequency $\nu_{en} = 1.489 \times 10^9$ Hz. In addition, from n_0 , we can compute the plasma frequency as $f_p = 4.903 \times 10^9$ Hz, and the plasma DC conductivity as $\sigma_{DC} = 28.21$ S/m. We briefly note that, in general, the electron-ion collision frequency ν_{ei} may also contribute to σ_{DC} , however, it can be neglected in our case. Our calculations (based on expressions in [27]) show that ν_{ei} is on the order of $\sim 10^6$ Hz.

IV. MODELING RESULTS

In order to compute the induced current in a single plasma column resulting from the interaction with the potential wave, the value of A_m^{ext} is needed. One of the external sources

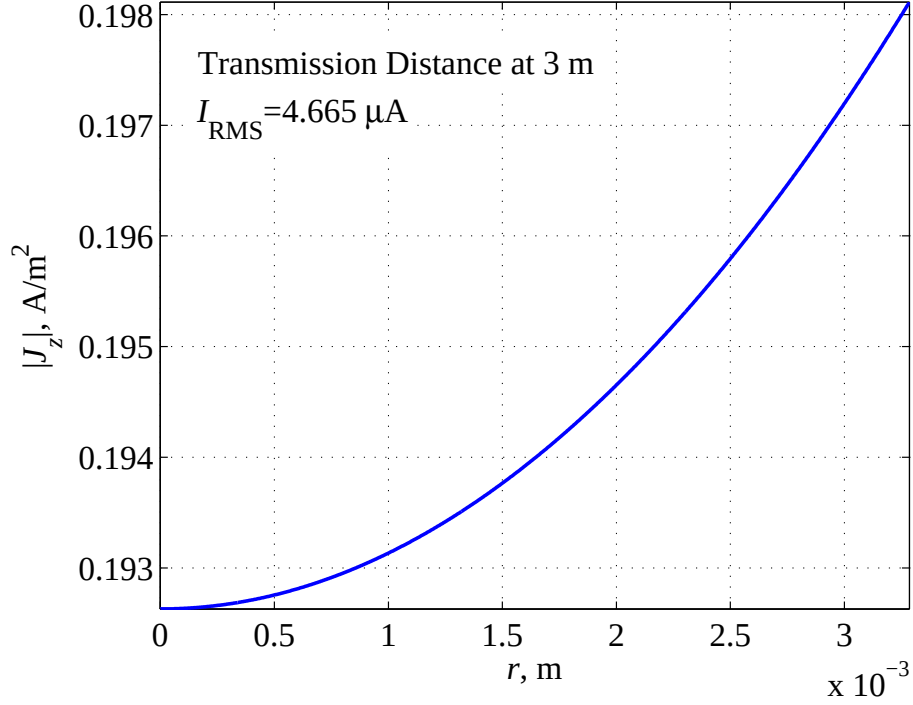


FIG. 4: The radial distribution of the magnitude of the z -directed induced RF current density in the fluorescent tube. Distance between transmitting dipole and receiving tube is 3 m and bias current is $I_0 = 200$ mA.

we use is a folded dipole which is aligned along the same axis as the fluorescent-tube detector, see Fig. 3. Its $\tilde{A}_z^{ext}$ potential component can be computed along the dipole's axis using Eq. (8). The magnitude of the driving current in the dipole I_m is calculated from the known antenna impedance ($\approx 73 \Omega$) and the power provided by the vector network analyzer (VNA) (set at 5 dBm). Note that the impedance of the VNA ports is 50Ω .

Fig. 4 illustrates the modeled distribution of the induced z -directed current density as a function of r where the distance between the transmitting dipole and the receiving fluorescent tube is 3 m and the bias current is $I_0 = 200$ mA. The current density is computed using Eq. (55). Note that the model uses the effective tube radius of $a_{\text{eff}} = 3.2854$ mm instead of the physical radius $a = 5$ mm. It is evident that the penetration depth in the low-temperature plasma is large. The RMS value of the induced current magnitude I_{RMS} shown in the plot is calculated using Eq. (58). For comparison, the RMS value of the total current calculated using Eq. (56) is $4.685 \mu\text{A}$. The small difference between the two is due to the fact that the

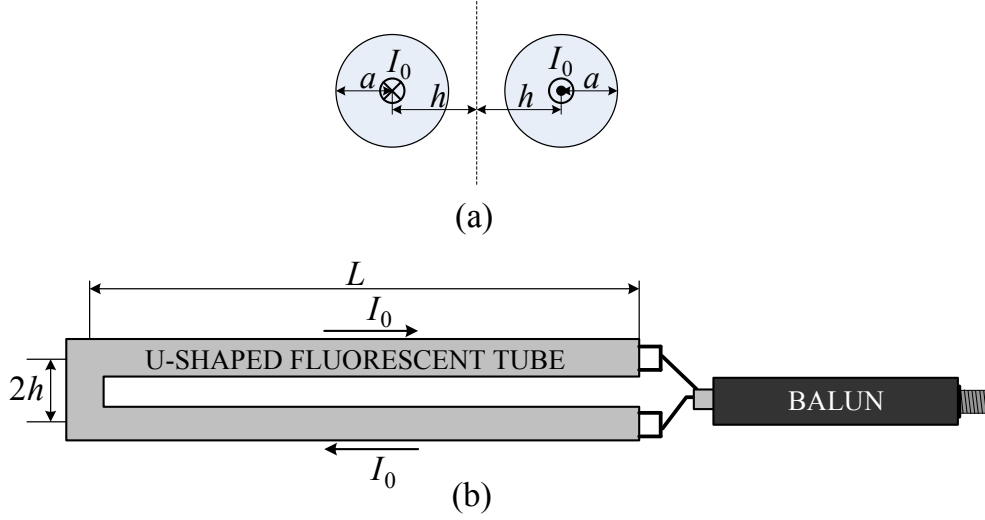


FIG. 5: Schematic of the U-tube detector: (a) cross-sectional view; (b) longitudinal view.

plasma conductivity is fairly low and the real part of p^2 is more than 10 times greater than its imaginary part. The situation is very different in a good conductor such as copper where the imaginary part of p^2 is three orders of magnitude greater than the real part. As a result, I_{RMS} in a biased copper cylinder would be practically zero.

From I_{RMS} , it is possible to estimate the magnitude of the transmission coefficient S_{21} in a measurement set-up with a VNA. Assuming impedance match, the power delivered to the $50\ \Omega$ load of the VNA is $50I_{\text{RMS}}^2$. For the case of 3 meters, the estimated S_{21} is about -65 dB, which is well above the noise level of the system and can be measured. To measure at larger distances where signals are weaker and cable loss is significant, an amplifier can be used, see Fig. 3.

V. DETECTION SCHEME AND EXPERIMENTAL SET-UP

We measure the transmission coefficient, i.e. the complex S_{21} parameter, between a dipole and a fluorescent-tube detector using an Advantest VNA R3765CG as shown in Fig 3. A $50\ \Omega$ coaxial cable connects the fluorescent tube to both its RF load (the VNA port) and the DC power supply. The DC and the RF paths are separated by a bias tee [33]. Fig. 5 shows a schematic of the U-shaped fluorescent tube used as a detector. Its plasma column forms a narrow rectangular loop of length $L = 13$ cm. The tube has a circular cross-section of radius

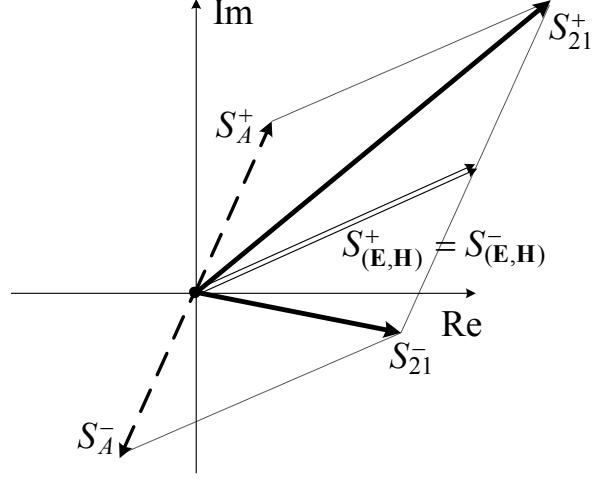


FIG. 6: Illustration of the potential and the power components of the complex S_{21} .

$a = 5$ mm. The distance between the centers of the two parallel tube arms is $2h = 15$ mm.

The dipole transmits considerable RF power azimuthally; however, axially, its RF power is ideally zero. Hereafter, we refer to the signals due to the conventional $(\mathbf{E}, \mathbf{H})$ radiation as *power* signals while the signals due to the potential alone are called *potential* signals. The dipole and the tube are aligned axially as shown schematically in Fig. 3 to closely match the condition of a longitudinally polarized vector-potential wave.

It is clear that, when ignited, the tube detector may act as a narrow loop and pick up interfering power signals. These may be picked up by the feed coaxial lines as well if the feed baluns are not well tuned. On the other hand, RF-power interference signals are inevitable due to non-ideal dipole-tube alignment and reflections. Thus, the detection scheme must include a method of eliminating them.

To achieve separation between the potential signal and the power signal, we make use of the fact that the phase of the potential signal depends on the polarity of the bias, $S_A^+ = -S_A^-$, while that of the power signal does not, $S_{(\mathbf{E}, \mathbf{H})}^+ = S_{(\mathbf{E}, \mathbf{H})}^-$. With the VNA, we have a phase-coherent system, which allows for the measurement of the complex transmission coefficient S_{21} . Thus, through a switch of the bias polarity we measure two S_{21} parameters, S_{21}^+ and S_{21}^- , from which the potential signal component S_A and the power signal component $S_{(\mathbf{E}, \mathbf{H})}$ can be extracted by simple subtraction and addition:

$$S_A = 0.5(S_{21}^+ - S_{21}^-), \quad S_{(\mathbf{E}, \mathbf{H})} = 0.5(S_{21}^+ + S_{21}^-). \quad (70)$$

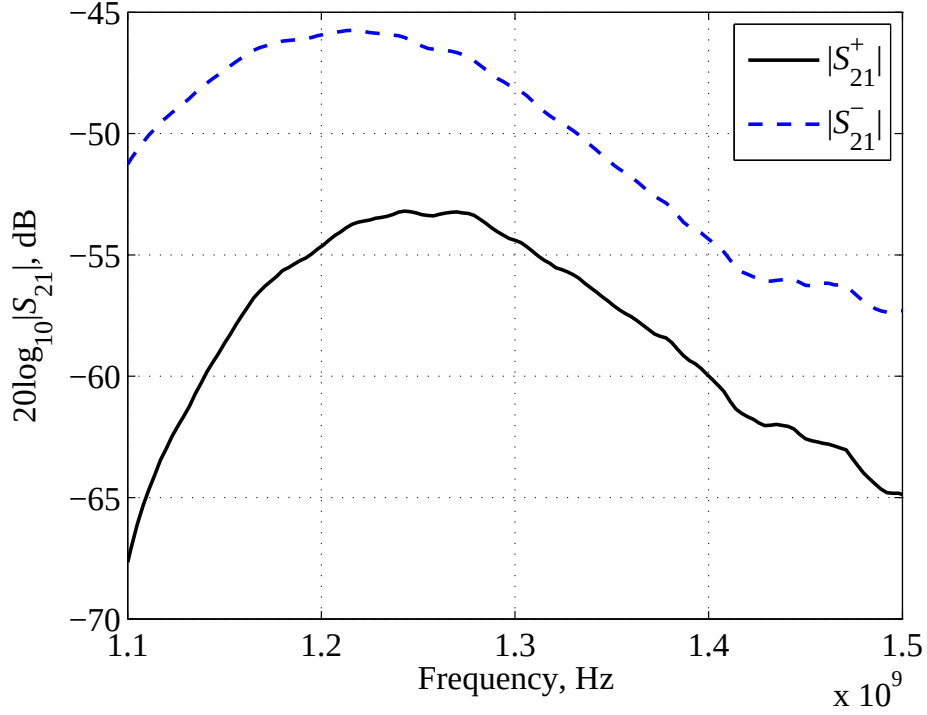


FIG. 7: The measured insertion loss at a distance $D = 1.2$ m with a positive and a negative DC bias of the fluorescent tube.

The technique is illustrated in Fig. 6.

VI. EXPERIMENTAL RESULTS

The frequency range of the measurements is between 1.1 GHz and 1.5 GHz where the tube length L is about half a wavelength. Also, these frequencies are well below the calculated plasma frequency of the low-temperature plasma. Two sets of measurements were performed. The first set is a measurement without an amplifier in a compact anechoic chamber, which allows for distances between the transmitting antenna and the tube detector between 1.2 m and 1.5 m. There, the transmitting antenna is a monopole whose return loss is minimum at 1.22 GHz and is about -16 dB. Fig. 7 and Fig. 8 show the magnitudes and phases of the measured S_{21} parameters at a distance of $D = 1.2$ m. The superscripts $+$ and $-$ indicate the assumed positive and negative polarity of the DC bias. Significant difference between the magnitudes of the two signals is observed.

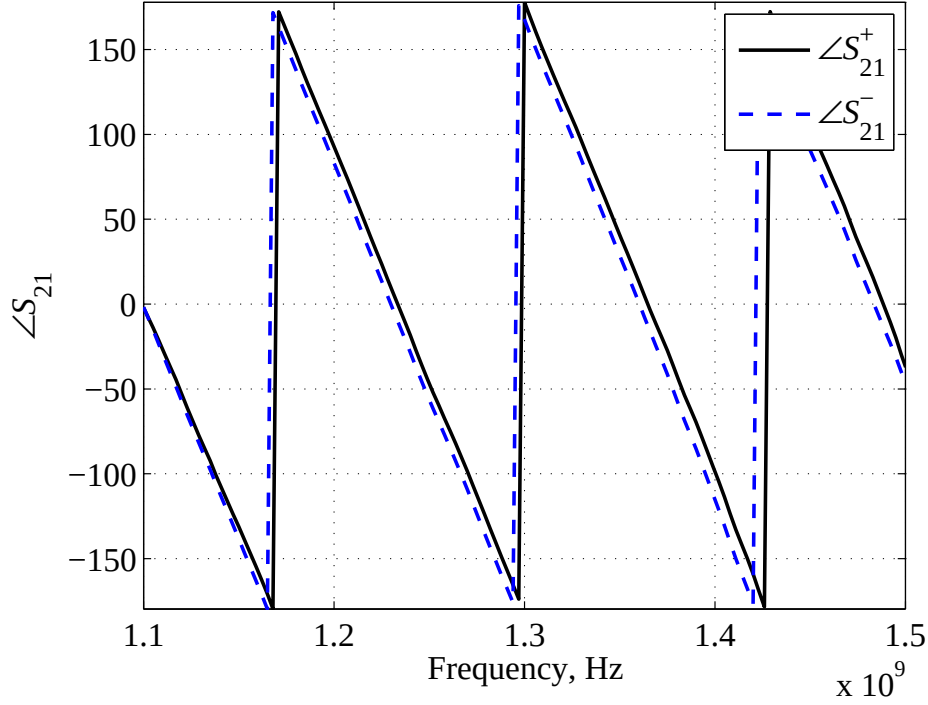


FIG. 8: The phase of the measured S_{21} parameter at a distance $D = 1.2$ m with a positive and a negative DC bias of the fluorescent tube.

The signals S_A and $S_{(\mathbf{E}, \mathbf{H})}$ are derived using Eq. (70). Their magnitudes (in dB) are plotted in Fig. 9. To illustrate the agreement with our theoretical model, we compute the magnitude of S_A at the center frequency of 1.3 GHz for a distance of 1.2 m. The obtained theoretical prediction of -56.7 dB is in good agreement with the measured value of -59.7 dB.

Another measure of the ability of the fluorescent tube detector to respond to the longitudinal potential wave can be defined as the ratio of the induced RMS current value and the RMS value of the incident vector potential:

$$R = \frac{I_{\text{RMS}}}{A_{\text{RMS}}^{\text{ext}}}, \quad (71)$$

where $A_{\text{RMS}}^{\text{ext}} = \frac{A_m^{\text{ext}}}{\sqrt{2}}$. We refer to R as the detector responsivity. The responsivity of the modeled BAYCO U-tube is calculated by our model as $264\,030 \text{ m} \times \text{H}^{-1}$ while the one corresponding to the measurements in the anechoic chamber is $186\,376 \text{ m} \times \text{H}^{-1}$. This is a factor of disagreement of 0.706. Such a disagreement is not surprising bearing in mind the

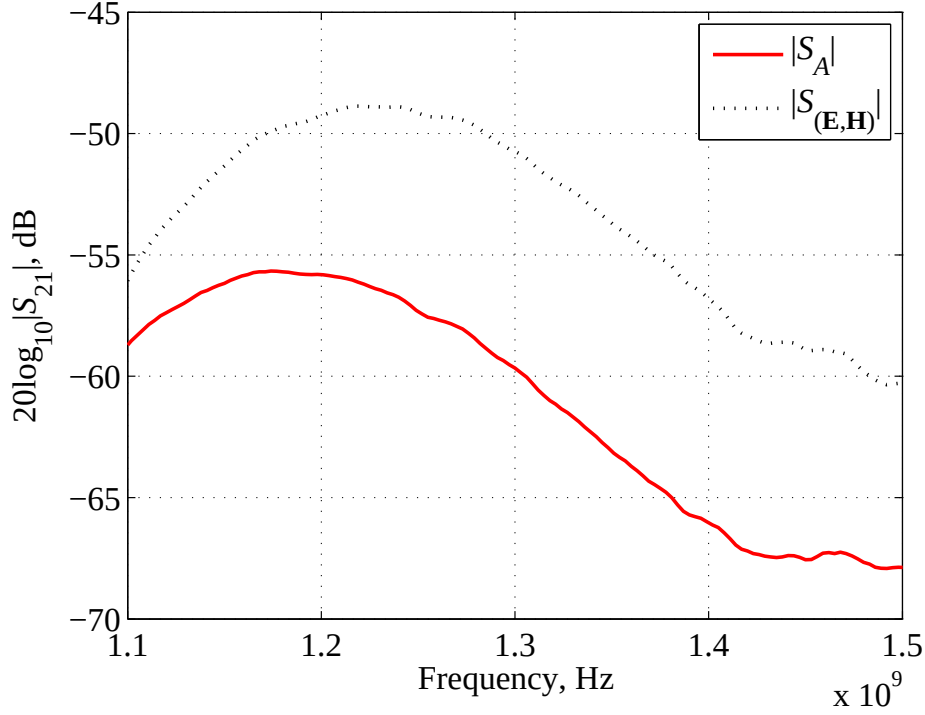


FIG. 9: Insertion loss for the potential signal S_A and the power signal $S_{(\mathbf{E},\mathbf{H})}$ at a distance $D = 1.2$ m derived from the measured complex transmission coefficients S_{21}^+ and S_{21}^- .

many approximations made in modeling the plasma parameters as well as the solution in the cylindrical volume.

In the second set of experiments, we measure the received S_A signal with respect to distance. This measurement was carried out in the open where the only significant reflections are due to the ground. The transmitting antenna is a folded dipole fed with a 4:1 balun. Its reflection loss is -15 dB at about 1.21 GHz. An amplifier was used, see Fig 3, to ensure high signal-to-noise ratio. The amplifier has a gain between 29 dB and 30 dB in the whole frequency band of interest. Data were taken every 25 cm. The comparison with the analytical prediction is shown in Fig. 10 for a height above ground $H = 1.26$ m. The analytical model takes into account the gain of the amplifier. Two analytical curves are shown—one shows the S_A response taking into account the direct-path signal only, while the other takes into account both the direct-path signal and the reflection from the ground. The reflection is modeled assuming perfect ground and using the method of images. The analytical responses are multiplied by the factor 0.706 before being plotted in Fig. 10. This

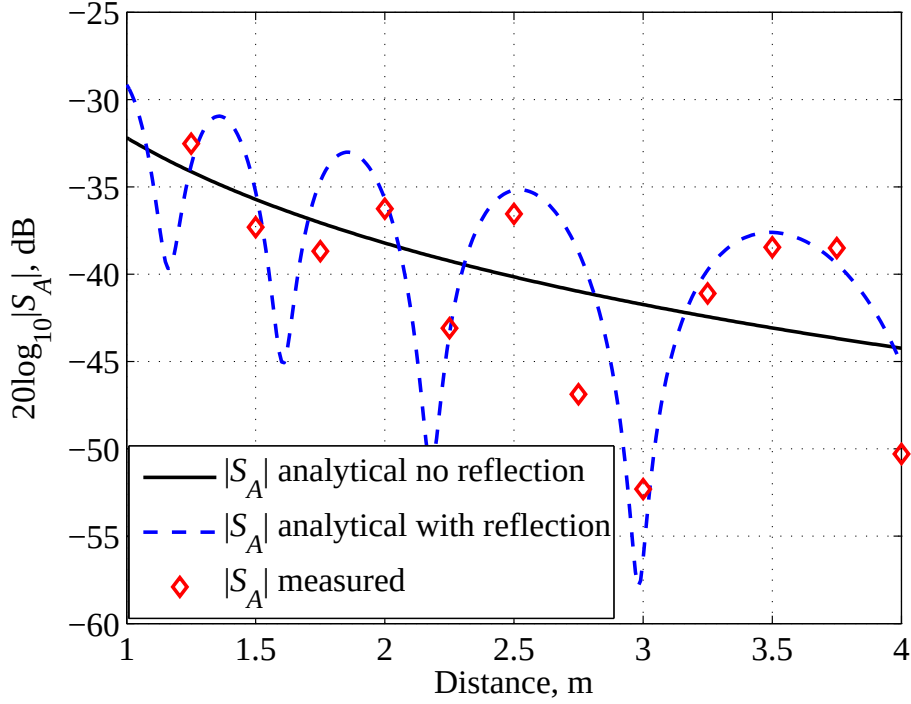


FIG. 10: The S_A signal *versus* distance. The solid curve shows the analytical calculation taking into account direct-path signal only. The dash curve shows analytical calculation taking into account both direct-path and reflected signals assuming perfect ground. Diamond markers show measurements. Analytical results are multiplied by 0.706.

factor of disagreement between the theoretical model and the measurements was obtained before through the measurements in the anechoic chamber.

VII. DISCUSSION AND CONCLUSIONS

We have presented results from a study of the measurability of the electromagnetic 4-vector potential at RF/microwave frequencies. They indicate it is possible to detect signals carried by the potential, which are independent from the $(\mathbf{E}, \mathbf{H})$ field. This is contrary to the opinion that the potential waves have no physical meaning.

We have presented results where the potential-wave detector uses a fluorescent tube. These results agree well with our theoretical predictions, which are based on the plasma parameters of Ar-Hg fluorescent tubes. We have carried out numerous experiments with

neon bulbs as well. The plasma parameters for small neon bulbs are not available to us and predictions are impossible at this stage. The experiments, however, suggest that neon bulbs (e.g., NE-2), despite their small size, have sufficient responsivity to obtain measurable potential signals.

The implications of our findings are far reaching. Particularly counter-intuitive is the fact that signal reception is possible via a potential wave which is not associated with electromagnetic power density. The potential-signal power is provided only locally by the DC bias of the detector. Further research is planned toward the confirmation of this result. Particularly important in this respect is the design of transmitting devices whose $(\mathbf{E}, \mathbf{H})$ field and, therefore, radiation power are significantly suppressed in all space. At the same time, they must be capable of generating a longitudinal potential wave in the desired directions. Unlike conventional radiators, such as dipoles and monopoles, these new transmitting devices must achieve near-zero radiation impedance, both in its real and imaginary parts. Complications arise from the fact that such devices would not be directly compatible with conventional RF/microwave measurement instrumentation which is based on a $50\ \Omega$ system impedance. Our further research is directed toward a system which can unambiguously prove that the received potential-signal power exceeds significantly the overall transmitter power.

Appendix

Taking the curl of both sides of Eq. (19) leads to a relation between the vector potential and the velocity as

$$-R_e \tilde{\mathbf{A}} = \eta \tilde{\mathbf{v}} + \frac{\nabla \tilde{f}}{j\omega}, \quad \eta = 1 + \frac{1}{j\omega\tau}, \quad (72)$$

where $\tilde{f}$ can be any scalar function of space. Substituting this result back in Eq. (19) shows that

$$-R_e \Phi = \boldsymbol{\nu} \cdot \tilde{\mathbf{v}} - \tilde{f}, \quad (73)$$

up to a function which is constant in space, so that its gradient vanishes. Such a function is of no interest in this particular solution since all field and velocity phasors are dependent at least on z as per the desired wave solution in the form $\exp(-jk_0 z)$.

The current density can now be expressed as

$$\tilde{\mathbf{J}} = \rho_0 \tilde{\mathbf{v}} = -\frac{\rho_0}{\eta} \left(R_e \tilde{\mathbf{A}} + \frac{\nabla \tilde{f}}{j\omega} \right). \quad (74)$$

The term proportional to $\nabla \tilde{f}$ is a nonradiating source whose electric field is confined within the source volume while its magnetic field is zero [16]. This field is [16]

$$\tilde{\mathbf{E}}^{nr} = -\frac{\tilde{\mathbf{J}}^{nr}}{j\omega\epsilon_0} = -\frac{\rho_0 \nabla \tilde{f}}{\omega^2 \epsilon_0 \eta}. \quad (75)$$

In the present problem, we consider an incident wave whose field components are locally zero. In particular, when $\tilde{E}_z^{ext} = 0$ at the boundary of the interaction volume, continuity requires that $\frac{\partial \tilde{f}}{\partial z}$ must vanish at the boundary. Since $\tilde{f}$ must be $\sim \exp(-jk_0 z)$, the latter is equivalent to $\tilde{f} = 0$ at the boundary. In addition, when $\tilde{E}_r^{ext} = 0$, $\frac{\partial \tilde{f}}{\partial r} = 0$ at the boundary. The two homogeneous boundary conditions ensure that $\tilde{f}$ must be identically zero everywhere in the cylindrical volume.

With $\tilde{f} = 0$, Eq. (72), Eq. (73) and Eq. (74) reduce to

$$-R_e \tilde{\mathbf{A}} = \eta \tilde{\mathbf{v}}, \quad (76)$$

$$-R_e \Phi = \boldsymbol{\nu} \cdot \tilde{\mathbf{v}}, \quad (77)$$

$$\tilde{\mathbf{J}} = -\frac{\rho_0 R_e}{\eta} \cdot \tilde{\mathbf{A}}. \quad (78)$$

Since the Lorenz gauge Eq. (20) must hold for the 4-potential, we find that

$$\eta \nabla \cdot \tilde{\mathbf{v}} = -\frac{j\omega}{c^2} (\boldsymbol{\nu} \cdot \tilde{\mathbf{v}}). \quad (79)$$

The substitution of Eq. (79) in the continuity equation

$$\nabla \cdot \tilde{\mathbf{J}} = \rho_0 \nabla \cdot \tilde{\mathbf{v}} = -j\omega \tilde{\rho} \quad (80)$$

leads to the expression for the charge density:

$$\tilde{\rho} = \rho_0 \frac{\boldsymbol{\nu} \cdot \tilde{\mathbf{v}}}{c^2 \eta}. \quad (81)$$

We briefly note that the arbitrary function $\tilde{f}$ should be set equal to zero also in the general case of external field values different from zero since the latter can be accounted for entirely by the boundary conditions for the 4-vector potential. Consequently, the boundary

conditions for the field of the nonradiating source would still be homogeneous. A formal proof of this general case, however, is not going to be pursued here.

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